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May 26, 2025

Desafio 2

Question 1.

Podemos escrever $\mathbf{X}'\mathbf{A}\mathbf{X}$ em termos dos desvios em relação à média:

$$\mathbf{X}'\mathbf{A}\mathbf{X} = (\mathbf{X} - \boldsymbol{\theta} + \boldsymbol{\theta})'\mathbf{A}(\mathbf{X} - \boldsymbol{\theta} + \boldsymbol{\theta}) = [(\mathbf{X} - \boldsymbol{\theta}) + \boldsymbol{\theta}]'\mathbf{A}[(\mathbf{X} - \boldsymbol{\theta}) + \boldsymbol{\theta}]$$

$$= \underbrace{(\mathbf{X} - \boldsymbol{\theta})'\mathbf{A}(\mathbf{X} - \boldsymbol{\theta})}_{1^{\text{o}}\text{ termo}} + \underbrace{(\mathbf{X} - \boldsymbol{\theta})'\mathbf{A}\boldsymbol{\theta}}_{2^{\text{o}}\text{ termo}} + \underbrace{\boldsymbol{\theta}'\mathbf{A}(\mathbf{X} - \boldsymbol{\theta})}_{3^{\text{o}}\text{ termo}} + \underbrace{\boldsymbol{\theta}'\mathbf{A}\boldsymbol{\theta}}_{4^{\text{o}}\text{ termo}} \tag{1}$$

No segundo termo, $(\mathbf{X} - \boldsymbol{\theta})'\mathbf{A}\boldsymbol{\theta}$ é um escalar:

$$(\mathbf{X} - \boldsymbol{\theta}) = \begin{bmatrix} x_1 - \theta_1 \\ x_2 - \theta_2 \\ \vdots \\ x_m - \theta_m \end{bmatrix}, \quad \mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1m} \\ a_{21} & a_{22} & \cdots & a_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mm} \end{bmatrix}, \quad \boldsymbol{\theta} = \begin{bmatrix} \theta_1 \\ \theta_2 \\ \vdots \\ \theta_m \end{bmatrix}$$

$$(\mathbf{X} - \boldsymbol{\theta})'\mathbf{A}\boldsymbol{\theta} = \sum_{k=1}^{m}(x_k - \theta_k)\sum_{l=1}^{m} a_{kl}\theta_l = \sum_{k=1}^{m}\sum_{l=1}^{m}(x_k - \theta_k)\, a_{kl}\, \theta_l$$

Um escalar é igual à sua transposta, portanto:

$$[(\mathbf{X} - \boldsymbol{\theta})'\mathbf{A}\boldsymbol{\theta}]' = \boldsymbol{\theta}'\mathbf{A}'(\mathbf{X} - \boldsymbol{\theta})$$

Como $\mathbf{A}$ é simétrica, temos que $\mathbf{A} = \mathbf{A}'$. Então:

$$\boldsymbol{\theta}'\mathbf{A}'(\mathbf{X} - \boldsymbol{\theta}) = \boldsymbol{\theta}'\mathbf{A}(\mathbf{X} - \boldsymbol{\theta}) \Rightarrow (\mathbf{X} - \boldsymbol{\theta})'\mathbf{A}\boldsymbol{\theta} = \boldsymbol{\theta}'\mathbf{A}(\mathbf{X} - \boldsymbol{\theta})$$

Portanto, o 2^{o} e 3^{o} termos são iguais, e assim, reescrevendo (1):

$$\begin{aligned}
\mathbf{X}'\mathbf{A}\mathbf{X} &= (\mathbf{X} - \boldsymbol{\theta})'\mathbf{A}(\mathbf{X} - \boldsymbol{\theta}) + \boldsymbol{\theta}'\mathbf{A}(\mathbf{X} - \boldsymbol{\theta}) + (\mathbf{X} - \boldsymbol{\theta})'\mathbf{A}\boldsymbol{\theta} + \boldsymbol{\theta}'\mathbf{A}\boldsymbol{\theta} \\
&= (\mathbf{X} - \boldsymbol{\theta})'\mathbf{A}(\mathbf{X} - \boldsymbol{\theta}) + 2\boldsymbol{\theta}'\mathbf{A}(\mathbf{X} - \boldsymbol{\theta}) + \boldsymbol{\theta}'\mathbf{A}\boldsymbol{\theta}
\end{aligned}$$

Cálculo da variância de $X'AX$:

$$\mathrm{Var}\boldsymbol{(X'AX)} = \mathbb{E}[\boldsymbol{(X'AX)^2}] - [\mathbb{E}\boldsymbol{(X'AX)}]^{\boldsymbol{2}}$$

1. Calculando $\mathbb{E}[\boldsymbol{(X'AX)}]^{\boldsymbol{2}}$

Temos que:

$$\begin{aligned}
\mathbb{E}[\mathbf{X}'\mathbf{A}\mathbf{X}] &= \mathrm{tr}(\mathbb{E}[\mathbf{X}'\mathbf{A}\mathbf{X}]) \\
&= \mathbb{E}[\mathrm{tr}(\mathbf{X}'\mathbf{A}\mathbf{X})] \\
&= \mathbb{E}[\mathrm{tr}(\mathbf{A}\mathbf{X}\mathbf{X}')] \\
&= \mathrm{tr}(\mathbb{E}[\mathbf{A}\mathbf{X}\mathbf{X}']) \\
&= \mathrm{tr}(\mathbf{A}\mathbb{E}[\mathbf{X}\mathbf{X}']) \\
&= \mathrm{tr}\left(\mathbf{A}\left(\mathrm{Var}[\mathbf{X}] + \boldsymbol{\theta}\boldsymbol{\theta}'\right)\right) \\
&= \mathrm{tr}(\mathbf{A}\boldsymbol{\Sigma}) + \mathrm{tr}(\mathbf{A}\boldsymbol{\theta}\boldsymbol{\theta}') \\
&= \mathrm{tr}(\mathbf{A}\mu_2\mathbf{I}) + \boldsymbol{\theta}'\mathbf{A}\boldsymbol{\theta} \\
&= \mu_2\,\mathrm{tr}(\mathbf{A}) + \boldsymbol{\theta}'\mathbf{A}\boldsymbol{\theta}
\end{aligned}$$

Logo,

$$\begin{aligned}\mathbb{E}[(\mathbf{X}'\mathbf{A}\mathbf{X})]^2 &= (\boldsymbol{\mu}_2 \operatorname{tr}(\mathbf{A}) + \boldsymbol{\theta}'\mathbf{A}\boldsymbol{\theta})^2 \\ &= \boldsymbol{\mu}_2^2 \operatorname{tr}(\mathbf{A})^2 + 2\boldsymbol{\mu}_2 \operatorname{tr}(\mathbf{A})\boldsymbol{\theta}'\mathbf{A}\boldsymbol{\theta} + (\boldsymbol{\theta}'\mathbf{A}\boldsymbol{\theta})^2\end{aligned}$$

2. Calculando $\mathbb{E}[\mathbf{(X'AX)^2}]$
Temos que:

$$\begin{aligned}(\mathbf{X}'\mathbf{A}\mathbf{X})^2 = &\left[(\mathbf{X}-\boldsymbol{\theta})'\mathbf{A}(\mathbf{X}-\boldsymbol{\theta})\right]^2 + 4\left[\boldsymbol{\theta}'\mathbf{A}(\mathbf{X}-\boldsymbol{\theta})\right]^2 + (\boldsymbol{\theta}'\mathbf{A}\boldsymbol{\theta})^2 \\ &+ 2\boldsymbol{\theta}'\mathbf{A}\boldsymbol{\theta}\left[(\mathbf{X}-\boldsymbol{\theta})'\mathbf{A}(\mathbf{X}-\boldsymbol{\theta}) + 4\boldsymbol{\theta}'\mathbf{A}\boldsymbol{\theta}\boldsymbol{\theta}'\mathbf{A}(\mathbf{X}-\boldsymbol{\theta})\right] \\ &+ 4\boldsymbol{\theta}'\mathbf{A}(\mathbf{X}-\boldsymbol{\theta})(\mathbf{X}-\boldsymbol{\theta})'\mathbf{A}(\mathbf{X}-\boldsymbol{\theta})\end{aligned}$$

Logo,

$$\begin{aligned}\mathbb{E}(\mathbf{X}'\mathbf{A}\mathbf{X})^2 = &\,\mathbb{E}\left[((\mathbf{X}-\boldsymbol{\theta})'\mathbf{A}(\mathbf{X}-\boldsymbol{\theta}))^2\right] + \mathbb{E}\left[4\,(\boldsymbol{\theta}'\mathbf{A}(\mathbf{X}-\boldsymbol{\theta}))^2\right] \\ &+ \mathbb{E}\left[(\boldsymbol{\theta}'\mathbf{A}\boldsymbol{\theta})^2\right] + \mathbb{E}\left[2\boldsymbol{\theta}'\mathbf{A}\boldsymbol{\theta}\cdot(\mathbf{X}-\boldsymbol{\theta})'\mathbf{A}(\mathbf{X}-\boldsymbol{\theta})\right] \\ &+ \mathbb{E}\left[4\boldsymbol{\theta}'\mathbf{A}\boldsymbol{\theta}\cdot\boldsymbol{\theta}'\mathbf{A}(\mathbf{X}-\boldsymbol{\theta})\right] \\ &+ \mathbb{E}\left[4\boldsymbol{\theta}'\mathbf{A}(\mathbf{X}-\boldsymbol{\theta})(\mathbf{X}-\boldsymbol{\theta})'\mathbf{A}(\mathbf{X}-\boldsymbol{\theta})\right]\end{aligned}$$

Assumindo $\mathbf{Y} = \mathbf{X} - \boldsymbol{\theta}$, e consequentemente, $\mathbb{E}[\mathbf{Y}] = \mathbf{0}$, temos que:

$$\begin{aligned}\mathbb{E}\left[(\mathbf{X}'\mathbf{A}\mathbf{X})^2\right] = &\,\mathbb{E}\left[(\mathbf{Y}'\mathbf{A}\mathbf{Y})^2\right] + 4\,\mathbb{E}\left[(\boldsymbol{\theta}'\mathbf{A}\mathbf{Y})^2\right] + (\boldsymbol{\theta}'\mathbf{A}\boldsymbol{\theta})^2 \\ &+ 2\,\boldsymbol{\theta}'\mathbf{A}\boldsymbol{\theta}\left(\mathbb{E}[\mathbf{Y}'\mathbf{A}\mathbf{Y}] + 4\,\boldsymbol{\theta}'\mathbf{A}\boldsymbol{\theta}\cdot\boldsymbol{\theta}'\mathbf{A}\,\mathbb{E}[\mathbf{Y}]\right) \\ &+ 4\,\mathbb{E}\left[\boldsymbol{\theta}'\mathbf{A}\mathbf{Y}\,\mathbf{Y}'\mathbf{A}\mathbf{Y}\right]\end{aligned}$$

Temos que, $\mathbb{E}[\mathbf{Y}'\mathbf{A}\mathbf{Y}] = \mu_2 \operatorname{tr}(\mathbf{A})$ e $\mathbb{E}[\mathbf{Y}] = \mathbf{0}$. Então

$$\mathbb{E}[(\mathbf{X}'\mathbf{A}\mathbf{X})^2] = \mathbb{E}[(\mathbf{Y}'\mathbf{A}\mathbf{Y})^2] + 4\mathbb{E}[(\boldsymbol{\theta}'\mathbf{A}\mathbf{Y})^2] + (\boldsymbol{\theta}'\mathbf{A}\boldsymbol{\theta})^2 + 2\boldsymbol{\theta}'\mathbf{A}\boldsymbol{\theta}\mu_2\operatorname{tr}(\mathbf{A}) + 4\mathbb{E}[\boldsymbol{\theta}'\mathbf{A}\mathbf{Y}\mathbf{Y}'\mathbf{A}\mathbf{Y}]$$

Vamos calcular cada termo da expressão acima:
- $\mathbb{E}[(\mathbf{Y}'\mathbf{A}\mathbf{Y})^2]$:
$(\mathbf{Y}'\mathbf{A}\mathbf{Y})^2$ pode ser escrito da seguinte maneira:

$$(\mathbf{Y}'\mathbf{A}\mathbf{Y})^2 = \left(\sum_{i,j} a_{ij}Y_iY_j\right)^2 = \sum_{i,j,k,l} a_{ij}a_{kl}Y_iY_jY_kY_l = \sum_i\sum_j\sum_k\sum_l a_{ij}a_{kl}Y_iY_jY_kY_l$$

$$\mathbb{E}\left[(\mathbf{Y}'\mathbf{A}\mathbf{Y})^2\right] = \sum_{i,j,k,l} a_{ij}a_{kl}\mathbb{E}(Y_iY_jY_kY_l)$$

Agora, precisamos calcular $\mathbb{E}(Y_iY_jY_kY_l)$ usando as hipóteses de independência e momentos centrados:

$$\mathbb{E}[Y_iY_jY_kY_l] = \begin{cases}\mu_4, & \text{se } i=j=k=l, \\ \mu_2^2, & \text{se formam dois pares iguais:} \\ & \quad (i=j \text{ e } k=l) \text{ ou } (i=k \text{ e } j=l) \text{ ou } (i=l \text{ e } j=k), \\ 0, & \text{caso contrário.}\end{cases}$$

Dessa forma:

1) **Caso** $i=j=k=l$:

$$\mathbb{E}\left[(\mathbf{Y}'\mathbf{A}\mathbf{Y})^2\right] = \sum_i a_{ii}^2\mu_4$$

2) **Caso** $i=j,\ k=l$ **(com** $k \neq i$**)**:

$$\mathbb{E}\left[(\mathbf{Y}'\mathbf{A}\mathbf{Y})^2\right] = \sum_i\sum_{k\neq i} a_{ii}a_{kk}\mu_2^2$$

3) **Caso** $i = k$**,** $j = l$ **(com** $j \neq i$**)**:

$$\mathbb{E}\left[(\mathbf{Y}'\mathbf{A}\mathbf{Y})^2\right] = \sum_i \sum_{j \neq i} a_{ij}^2 \mu_2^2$$

4) **Caso** $i = l$**,** $j = k$ **(com** $j \neq i$**)**:

$$\mathbb{E}\left[(\mathbf{Y}'\mathbf{A}\mathbf{Y})^2\right] = \sum_i \sum_{j \neq i} a_{ij} a_{ji} \mu_2^2$$

Portanto, somando todos os casos temos que:

$$\mathbb{E}[(\mathbf{Y}'\mathbf{A}\mathbf{Y})^2] = \mu_4 \sum_i a_{ii}^2 + \mu_2^2 \left[\sum_i \sum_{k \neq i} a_{ii} a_{kk} + \sum_i \sum_{j \neq i} a_{ij}^2 + \sum_i \sum_{j \neq i} a_{ij} a_{ji} \right]$$

Note, que como $\mathbf{A}$ é simétrica, vale que $a_{ij} = a_{ji}$, então:

$$\sum_i \sum_{j \neq i} a_{ij} a_{ji} = \sum_i \sum_{j \neq i} a_{ij}^2 = \mathrm{tr}(\mathbf{A}^2)$$

Temos também que:

$$\sum_i \sum_{k \neq i} a_{ii} a_{kk} = \sum_i a_{ii} \sum_{k \neq i} a_{kk} = \mathrm{tr}(\mathbf{A}) \cdot \mathrm{tr}(\mathbf{A}) = [\mathrm{tr}(\mathbf{A})]^2$$

Além disso, a expressão $\mu_4 \sum_i a_{ii}^2$ pode ser reescrita como $(\mu_4 - 3\mu_2^2)\mathbf{a}'\mathbf{a}$, pois:
Dessa forma,

$$\mathbb{E}\left[(\mathbf{Y}'\mathbf{A}\mathbf{Y})^2\right] = (\mu_4 - 3\mu_2^2)\, \mathbf{a}'\mathbf{a} + \mu_2^2 \left[\mathrm{tr}(\mathbf{A})^2 + 2\ \mathrm{tr}(\mathbf{A}^2)\right]$$

- $4\mathbb{E}\left[(\boldsymbol{\theta}'\mathbf{A}\mathbf{Y})^2\right]$:
Considerando $\mathbf{b} = \mathbf{A}\boldsymbol{\theta}$, teremos que:

$$(\boldsymbol{\theta}'\mathbf{A}\mathbf{Y})^2 = (\mathbf{A}\boldsymbol{\theta})'\mathbf{Y})^2 = (\mathbf{b}'\mathbf{Y})^2 = \left(\sum_i \mathbf{b}_i \mathbf{Y}_i\right)^2 = \sum_i \sum_j \mathbf{b}_i \mathbf{b}_j \mathbf{Y}_i \mathbf{Y}_j$$

Dessa forma,

$$\mathbb{E}\left[(\boldsymbol{\theta}'\mathbf{A}\mathbf{Y})^2\right] = \sum_i \sum_j \mathbf{b}_i \mathbf{b}_j\, \mathbb{E}(\mathbf{Y}_i \mathbf{Y}_j)$$

Com relação a $\mathbb{E}(\mathbf{Y}_i \mathbf{Y}_j)$:
Sabendo que $\mu_r = \mathbb{E}[(X_i - \theta_i)^r]$, logo temos para $\mathbf{Y}$ que

$$\mu_r = \mathbb{E}[(\mathbf{Y}_i + \boldsymbol{\theta}_i - \boldsymbol{\theta}_i)^r] = \mathbb{E}[(\mathbf{Y}_i)^r]$$

- Se $i = j$, então $\mathbb{E}(\mathbf{Y}_i \mathbf{Y}_j) = \mathbb{E}(\mathbf{Y}_i^2) = \mu_2$
- Se $i \neq j$, então $\mathbb{E}(\mathbf{Y}_i \mathbf{Y}_j) = 0$

Assim, somando todos os casos:

$$4\,\mathbb{E}\left[(\boldsymbol{\theta}'\mathbf{A}\mathbf{Y})^2\right] = 4 \sum_i \mathbf{b}_i^2 \mu_2 + 0 = 4\mu_2 \sum_i \mathbf{b}_i^2 = 4\mu_2\, \mathbf{b}'\mathbf{b} = 4\mu_2\, (\mathbf{A}\boldsymbol{\theta})'(\mathbf{A}\boldsymbol{\theta}) = 4\mu_2\, \boldsymbol{\theta}'\mathbf{A}'\mathbf{A}\boldsymbol{\theta} = 4\mu_2\, \boldsymbol{\theta}'\mathbf{A}^2\boldsymbol{\theta}$$

- $4\mathbb{E}[\boldsymbol{\theta}'\mathbf{A}\mathbf{Y}\mathbf{Y}'\mathbf{A}\mathbf{Y}]$:
Assumindo $\mathbf{b} = \mathbf{A}\boldsymbol{\theta}$, temos que:

$$\boldsymbol{\theta}'\mathbf{A}\mathbf{Y}\mathbf{Y}'\mathbf{A}\mathbf{Y} = \sum_i \sum_j \sum_k \mathbf{b}_i \mathbf{a}_{jk} \mathbf{Y}_i \mathbf{Y}_j \mathbf{Y}_k$$

Logo,

$$\mathbb{E}(\boldsymbol{\theta}'\mathbf{A}\mathbf{Y}\mathbf{Y}'\mathbf{A}\mathbf{Y}) = \sum_i \sum_j \sum_k \mathbf{b}_i \mathbf{a}_{jk} \mathbb{E}(\mathbf{Y}_i \mathbf{Y}_j \mathbf{Y}_k)$$

Se $i = j = k$, então $\mathbb{E}(\mathbf{Y}_i \mathbf{Y}_j \mathbf{Y}_k) = \mathbb{E}(\mathbf{Y}_i^3) = \mu_3$
Se entre i, j e k houver pelo menos um par diferente, então $\mathbb{E}(\mathbf{Y}_i \mathbf{Y}_j \mathbf{Y}_k) = 0$

Assim,

$$4\,\mathbb{E}[\boldsymbol{\theta}'\mathbf{A}\mathbf{Y}\mathbf{Y}'\mathbf{A}\mathbf{Y}] = 4\sum_i \mathbf{b}_i\mathbf{a}_{ii}\mu_3 = 4\mu_3\sum_i \mathbf{b}_i\mathbf{a}_{ii} = 4\mu_3\,\mathbf{b}'\mathbf{a} = 4\mu_3\,\boldsymbol{\theta}'\mathbf{A}\mathbf{a}$$

Portanto,

$$\begin{aligned}
\mathbb{E}[(\mathbf{X}'\mathbf{A}\mathbf{X})^2] &= \mathbb{E}[(\mathbf{Y}'\mathbf{A}\mathbf{Y})^2] + 4\,\mathbb{E}[(\boldsymbol{\theta}'\mathbf{A}\mathbf{Y})^2] + (\boldsymbol{\theta}'\mathbf{A}\boldsymbol{\theta})^2 \\
&\quad + 2\,\boldsymbol{\theta}'\mathbf{A}\boldsymbol{\theta}\,\mathbb{E}(\mathbf{Y}'\mathbf{A}\mathbf{Y}) + 4\mu_3\,\boldsymbol{\theta}'\mathbf{A}\mathbf{a} \\
&= (\mu_4 - 3\mu_2^2)\,\mathbf{a}'\mathbf{a} + \mu_2^2[\mathrm{tr}(\mathbf{A})^2 + 2\ \mathrm{tr}(\mathbf{A}^2)] \\
&\quad + 4\mu_2\,\boldsymbol{\theta}'\mathbf{A}^2\boldsymbol{\theta} + (\boldsymbol{\theta}'\mathbf{A}\boldsymbol{\theta})^2 + 2\mu_2\,\boldsymbol{\theta}'\mathbf{A}\boldsymbol{\theta}\ \mathrm{tr}(\mathbf{A}) + 4\mu_3\,\boldsymbol{\theta}'\mathbf{A}\mathbf{a}
\end{aligned}$$

Por fim, voltando à fórmula da variância, substituindo os valores, temos:

$$\begin{aligned}
\mathrm{Var}(X'AX) &= \mathbb{E}[(\mathbf{X}'\mathbf{A}\mathbf{X})^2] - [\mathbb{E}(\mathbf{X}'\mathbf{A}\mathbf{X})]^2 \\
&= \Big[(\mu_4 - 3\mu_2^2)\mathbf{a}'\mathbf{a} + \mu_2^2\Big(\ \mathrm{tr}(\mathbf{A})^2 + 2\ \mathrm{tr}(\mathbf{A}^2)\Big) + 4\mu_2\boldsymbol{\theta}'\mathbf{A}^2\boldsymbol{\theta} + (\boldsymbol{\theta}'\mathbf{A}\boldsymbol{\theta})^2 \\
&\quad + 2\mu_2\boldsymbol{\theta}'\mathbf{A}\boldsymbol{\theta}\,\mathrm{tr}(\mathbf{A}) + 4\mu_3\boldsymbol{\theta}'\mathbf{A}\mathbf{a}\Big] \\
&\quad - \Big[\mu_2^2\,\mathrm{tr}(\mathbf{A})^2 + 2\mu_2\,\mathrm{tr}(\mathbf{A})\boldsymbol{\theta}'\mathbf{A}\boldsymbol{\theta} + (\boldsymbol{\theta}'\mathbf{A}\boldsymbol{\theta})^2\Big] \\
&= (\mu_4 - 3\mu_2^2)\mathbf{a}'\mathbf{a} + 2\mu_2^2\,\mathrm{tr}(\mathbf{A}^2) + 4\mu_2\boldsymbol{\theta}'\mathbf{A}^2\boldsymbol{\theta} + 4\mu_3\boldsymbol{\theta}'\mathbf{A}\mathbf{a}
\end{aligned}$$

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